

Mid-semester Exam - Deferred

EGH445 Modern Control - 2025

Instructions:

This quiz has a total of 3 problems. The total number of possible marks for this quiz is 100. It is recommended that you attempt to answer all questions. **All answers need to be carefully justified.** Explanations of the steps used are essential, and will be carefully accounted for in the assessment. Best marks are awarded to fully justified and numerically correct solutions, but partial credits may be given to partially complete solutions. Any assumption made should be consistent with the question, clearly stated and justified.

Problem 1

Consider the phase plots for two different systems each with states $x(t)$ and $y(t)$ in Figure 1.

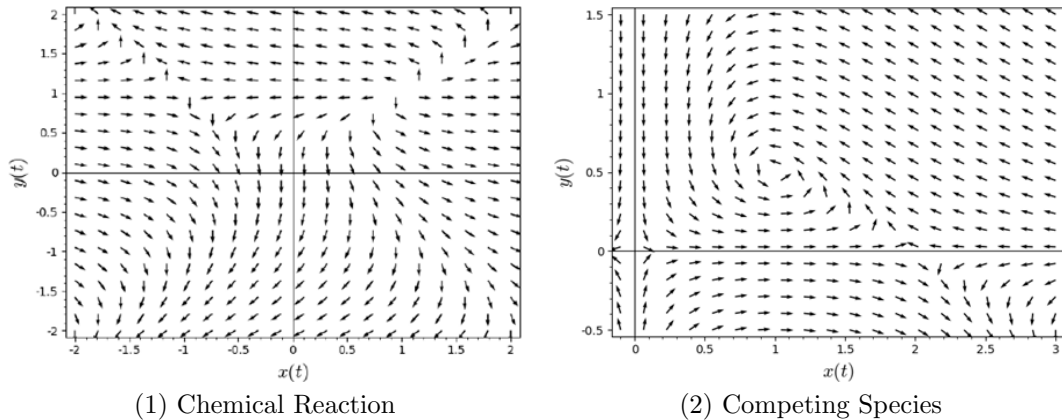


Figure 1: Phase Portraits.

- A) Is the chemical reaction in system (1) stable? Justify your answer. (6/100 marks)
- B) Identify the equilibrium points of system (2) and state whether each point is a stable, unstable or saddle equilibrium point of the system. (9/100 marks)
- C) If the population of species in system (2) was initially equal (i.e. $x(0) = y(0)$) how would you expect their populations to change? State any assumptions. (5/100 marks)
- D) For system (2) how would you determine if the species populations can be controlled using state-feedback and assuming only $x(t)$ can be measured or sensed? (10/100 marks)

Problem 2

A non-linear system is described by the following state equations:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.5x_1 - \sin(x_2) + u\end{aligned}$$

- A) Find the equilibrium state $(\bar{x}_1, \bar{x}_2)$ corresponding to a constant input $u = \bar{u} = 1$.
(10/100 marks)
- B) Linearise the system dynamics (state equations) around this equilibrium point $(\bar{x}_1, \bar{x}_2, \bar{u})$. Express the result in the standard linear state-space form $\delta\dot{x} = A\delta x + B\delta u$. Clearly state the matrices A and B .
(10/100 marks)
- C) Discretise the system assuming a sampling time $T = 1$ s and analyse the stability of the equilibrium point.
(10/100 marks)

Problem 3

A discrete-time system subject to a constant disturbance $d(kT)$ is given by:

$$\begin{aligned}x(kT + T) &= 0.7x(kT) + u(kT) + 0.5d(kT) \\ y(kT) &= x(kT)\end{aligned}$$

The objective is to design a controller such that the output $y(kT)$ tracks a constant reference $r(kT)$ with zero steady-state error, while rejecting the constant disturbance $d(kT)$.

- A) Design a state feedback controller that achieves zero steady-state error. Select closed-loop poles such that the settling time is 5 seconds (*Hint: assume $T_s = 4/\sigma$ where σ is the real part of the (continuous-time) dominant pole*).
(30/100 marks)
- B) Explain if this controller structure achieves zero steady-state error for the constant reference $r(kT)$ when subject to a constant disturbance $d(kT)$. If your proposed controller does not reject a constant disturbance, briefly explain what would be the process to design a controller that does reject such disturbance.
(10/100 marks)

END OF QUIZ

RESOURCE SHEET

	$X(s)$	$x(t)$	$x(kT)$ or $x(k)$	$X(z)$
1.	—	—	Kronecker delta $\delta_0(k)$ 1 $k = 0$ 0 $k \neq 0$	1
2.	—	—	$\delta_0(n-k)$ 1 $n = k$ 0 $n \neq k$	z^{-k}
3.	$\frac{1}{s}$	$1(t)$	$1(k)$	$\frac{1}{1-z^{-1}}$
4.	$\frac{1}{s+a}$	e^{-at}	e^{-akT}	$\frac{1}{1-e^{-aT}z^{-1}}$
5.	$\frac{1}{s^2}$	t	kT	$\frac{Tz^{-1}}{(1-z^{-1})^2}$
6.	$\frac{2}{s^3}$	t^2	$(kT)^2$	$\frac{T^2 z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
7.	$\frac{6}{s^4}$	t^3	$(kT)^3$	$\frac{T^3 z^{-1}(1+4z^{-1}+z^{-2})}{(1-z^{-1})^4}$
8.	$\frac{a}{s(s+a)}$	$1 - e^{-at}$	$1 - e^{-akT}$	$\frac{(1-e^{-aT})z^{-1}}{(1-z^{-1})(1-e^{-aT}z^{-1})}$
9.	$\frac{b-a}{(s+a)(s+b)}$	$e^{-at} - e^{-bt}$	$e^{-akT} - e^{-bkT}$	$\frac{(e^{-aT} - e^{-bT})z^{-1}}{(1-e^{-aT}z^{-1})(1-e^{-bT}z^{-1})}$
10.	$\frac{1}{(s+a)^2}$	te^{-at}	kTe^{-akT}	$\frac{Te^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
11.	$\frac{s}{(s+a)^2}$	$(1-at)e^{-at}$	$(1-akT)e^{-akT}$	$\frac{1-(1+aT)e^{-aT}z^{-1}}{(1-e^{-aT}z^{-1})^2}$
12.	$\frac{2}{(s+a)^3}$	$t^2 e^{-at}$	$(kT)^2 e^{-akT}$	$\frac{T^2 e^{-aT}(1+e^{-aT}z^{-1})z^{-1}}{(1-e^{-aT}z^{-1})^3}$
13.	$\frac{a^2}{s^2(s+a)}$	$at - 1 + e^{-at}$	$akT - 1 + e^{-akT}$	$\frac{[(aT-1+e^{-aT})+(1-e^{-aT}-aTe^{-aT})z^{-1}]z^{-1}}{(1-z^{-1})^2(1-e^{-aT}z^{-1})}$
14.	$\frac{\omega}{s^2+\omega^2}$	$\sin \omega t$	$\sin \omega kT$	$\frac{z^{-1} \sin \omega T}{1-2z^{-1} \cos \omega T + z^{-2}}$
15.	$\frac{s}{s^2+\omega^2}$	$\cos \omega t$	$\cos \omega kT$	$\frac{1-z^{-1} \cos \omega T}{1-2z^{-1} \cos \omega T + z^{-2}}$
16.	$\frac{\omega}{(s+a)^2+\omega^2}$	$e^{-at} \sin \omega t$	$e^{-akT} \sin \omega kT$	$\frac{e^{-aT} z^{-1} \sin \omega T}{1-2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
17.	$\frac{s+a}{(s+a)^2+\omega^2}$	$e^{-at} \cos \omega t$	$e^{-akT} \cos \omega kT$	$\frac{1-e^{-aT} z^{-1} \cos \omega T}{1-2e^{-aT} z^{-1} \cos \omega T + e^{-2aT} z^{-2}}$
18.	—	—	a^k	$\frac{1}{1-az^{-1}}$
19.	—	—	a^{k-l} $k = 1, 2, 3, \dots$	$\frac{z^{-1}}{1-az^{-1}}$
20.	—	—	ka^{k-l}	$\frac{z^{-1}}{(1-az^{-1})^2}$
21.	—	—	$k^2 a^{k-l}$	$\frac{z^{-1}(1+az^{-1})}{(1-az^{-1})^3}$
22.	—	—	$k^3 a^{k-l}$	$\frac{z^{-1}(1+4az^{-1}+a^2 z^{-2})}{(1-az^{-1})^4}$
23.	—	—	$k^4 a^{k-l}$	$\frac{z^{-1}(1+11az^{-1}+11a^2 z^{-2}+a^3 z^{-3})}{(1-az^{-1})^5}$
24.	—	—	$a^k \cos k\pi$	$\frac{1}{1+az^{-1}}$

$x(t) = 0$ for $t < 0$
 $x(kT) = x(k) = 0$ for $k < 0$
 Unless otherwise noted, $k = 0, 1, 2, 3, \dots$